

Jonathan Tielve

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Education

University of Central Florida

May 2025

Minor, Computer Science. Bachelors, General Studies

Relevant Coursework: Computer Science 1 and 2, UCF Foundation Exam, Algorithms for Machine Learning, Computer Architecture.

Projects

CubeSat Attitude Control Testbed (C | Embedded Hardware Integration | STM32CubeMX | Sensor Fusion | FreeRTOS)

- Designed and integrated a 3-axis reaction wheel attitude control system on a 1.5U scale CubeSat using TB6612FNG motor drivers and 3 brushed DC motors.
- Developed firmware on a STM32H743 development kit featuring a Multiplicative Extended Kalman Filter for sensor fusion and attitude estimation.
- Integrated AS5600 magnetic encoders to create a closed-loop system for more precise PID control in conjunction with a 9-DOF IMU for accurate and quick attitude control.
- Used Webots to simulate control loop and acquire accurate biases for Kalman Filter.
- Performed hardware bring-up and wiring of all subsystems, including motor drivers and sensors ensuring noise isolation and stable operation.

Fixed-point Q15/Q7 Neural Network Library (C | STM32 | 2D Convolution | Forward Propagation | Activations)

- Developed a bare-metal neural network library for the STM32H743 using Q15 and Q7 fixed-point arithmetic, cutting RAM usage by up to 75%.
- Implemented full forward propagation for dense layers, 2D convolution, activation functions (ReLU, TanH, Sigmoid, Leaky ReLU), and regularization techniques (Dropout, Batch/Layer Normalization).
- Built with modularity in mind with separate activation, back propagation, computation, loss and regularization headers.

Maze Solving ESP32 (C | ESP32 | Zephyr | Reinforcement learning)

- Implemented Epsilon-Greedy-Q-Learning on an ESP32 for deeper learning, rewarding the agent for getting to the goal.
- Created a random maze generator using Depth-First search to guarantee solvable mazes.
- Utilized Zephyr Semaphores to allow the agent to wait on simulated sensor data.

Hardware Abstraction Layer for STM32H743 (C | Bare-Metal | Registers | RCC)

- Designed a Hardware Abstraction Layer (HAL) for the STM32H743 using a handle-based API facilitating clean abstraction.
- Implemented UART, SPI, GPIO, RCC and SysTick drivers, and configured clock tree for peripheral clocks.

CI/CD Home Server (Linux | Jenkins | Git)

- Created a home server for the purpose of hosting my own applications, designed with security in mind.
- Integrated Jenkins and Docker to host applications automatically after pushing to Gitea.
- Implemented a PfSense firewall separates WAN and LAN for added security.

Group Experience

University of Central Florida

Fall 2023

Technical Writer for UI/UX Class

- Collaborated with two frontend developers, a project lead, and a business analyst to create a brand new website for a Vietnamese restaurant to increase customers.
- Led documentation for cross-function project team, delivering key deliverables to the client.
- Wrote user friendly technical documentation for the restaurant's staff to be able to edit the menu or other elements.

Technical Skills

Programming Languages: C, C++, Python, Rust.

Embedded Systems: STM32 development, STM32CubeIDE, STM32CubeMX, Bare metal programming, Register manipulation, FreeRTOS, Zephyr, KiCad, PCB Design.

Machine Learning: Fixed-point arithmetic, Neural Network design, Embedded ML optimization, PyTorch, TensorFlow, scikit-learn.

Development Tools: Docker, Kubernetes, Jenkins, Git, REST API, MySQL, SAST, DAST.